

Learning to Use Sparse Native-KV Memory: A Controlled Adaptation Ladder for Progressive Retrieval Attention

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Abstract

Progressive Retrieval Attention (PRA) can route to external references and materialize their layer-native key/value (K/V) state, yet a transformer trained only with ordinary self-attention need not know how to use that state. We isolate this consumer-learning problem in a controlled associative-chain model. Reference identities are fixed by an oracle, reference K/V remains inside the autograd graph, and memory joins local K/V under the model’s ordinary attention softmax. Starting from one matched LocalSA checkpoint per seed, we compare Frozen PRA, three LoRA scopes, full-weight continued training, and PRA-native training from scratch. Each regime is evaluated with no memory, matched distractors, exact evidence, and the whole parent. Across 5 paired seeds, the best change in the correct-memory margin effect over Frozen is 9.512 for Native scratch. Evidence-only accuracy rises from 0.303 Frozen to 0.821 with Consumer LoRA and 0.992 from scratch, while whole-parent accuracy plateaus near one half. The controlled gate passes 3 preregistered criteria and yields the decision `pass_to_tier1`. These results concern oracle-selected memory consumption in a sub-million-parameter diagnostic model. They do not establish learned routing, language-model scaling, or deployment efficiency.

1 Introduction

Retrieval-augmented systems are often evaluated as if selecting the right text were sufficient. Selection is only one boundary. The downstream transformer must also turn retrieved state into a useful residual update without being distracted by irrelevant state. Architectures trained with retrieval in the loop, including REALM, RAG, and RETRO, make this interface part of learning [4, 6, 2]. A frozen transformer retrofitted with a new memory path has no such guarantee.

PRA makes this distinction unusually visible. It stores historical token K/V in the same representation used by a transformer’s attention layers, while compact gists are only an address index. Once a reference is selected, full token K/V is placed before local K/V under a joint softmax. This avoids a separate cross-attention value space, but native format does not imply native use. Queries, memory keys, values, output projections, and downstream feed-forward layers were all optimized for a different context topology.

This paper asks a deliberately narrower question before spending compute on a larger language model:

When memory identity and physical K/V are correct, how much transformer plasticity is needed before sparse memory improves the answer?

The experiment is a causal gate, not a benchmark race. We reuse randomized associative chains, hold the PRA topology fixed, and change only adaptation capacity. The design follows a strict

ladder from no trainable parameters through targeted low-rank updates [5] to full and from-scratch training. Four memory interventions separate useful state from the generic effect of adding tokens. Layerwise attention and representation-portability measurements then show where any output change enters the network.

Contributions. We provide (i) a differentiable native-KV training path that preserves the same joint attention operation used at inference; (ii) a matched six-rung PRA adaptation ladder plus GlobalSA and LocalSA controls; (iii) causal metrics linking memory identity to attention, residual state, answer margin, and prediction; and (iv) a stage-gated result that explicitly decides whether larger-model work is warranted.

2 Why a Native Interface May Still Be Foreign

Consider a decoder layer ℓ with local hidden state $H_L^{(\ell)}$ and a separately encoded memory sequence $H_M^{(\ell)}$. PRA reuses the layer’s ordinary projections:

$$Q_L^{(\ell)} = W_Q^{(\ell)} \text{LN}(H_L^{(\ell)}), \quad (1)$$

$$K_M^{(\ell)}, V_M^{(\ell)} = W_{K,V}^{(\ell)} \text{LN}(H_M^{(\ell)}), \quad (2)$$

$$K_L^{(\ell)}, V_L^{(\ell)} = W_{K,V}^{(\ell)} \text{LN}(H_L^{(\ell)}). \quad (3)$$

After the configured positional transform, memory and local keys share one normalizer:

$$A^{(\ell)} = \text{softmax} \left(\frac{Q_L^{(\ell)} [K_M^{(\ell)}; K_L^{(\ell)}]^\top}{\sqrt{d_h}} + \mathcal{M}_{\text{causal}} \right), \quad U^{(\ell)} = W_O^{(\ell)} A^{(\ell)} [V_M^{(\ell)}; V_L^{(\ell)}]. \quad (4)$$

Memory positions are visible to every local query; the local block remains causal.

This operation introduces no cross-attention projection. Nevertheless, four mismatches can remain. A query may not score remote keys correctly. Memory keys may depend too strongly on the context in which they were encoded. Memory values may point in directions the downstream residual stream does not interpret. Later layers may erase a useful update. Frozen evaluation conflates all four. The adaptation ladder opens them in a controlled order.

3 Differentiable Oracle-Memory Training

Offline PRA caches detach reference state so one encoding can serve many queries. That is the correct inference boundary but the wrong training boundary. During Gate 0, the selected reference tokens are encoded in the same forward graph as the query. Immediately before each PRA consumer layer, the model captures $K_M^{(\ell)}, V_M^{(\ell)}$ with detachment disabled. The query then consumes those tensors through the native joint softmax above. Gradients can reach Q/K/V/O, reference production, and subsequent consumers according to the selected regime.

```
memory_h = embed(reference_tokens)
for layer in decoder:
    if layer is PRA:
        memory_kv[layer] = project_kv(LN(memory_h), detach=False)
        memory_h = layer.local_forward(memory_h)
```

```

query_h = embed(query_tokens)
for layer in decoder:
    if layer is PRA:
        update = native_joint_attention(LN(query_h), memory_kv[layer])
        query_h = query_h + update
        query_h = query_h + FFN(LN(query_h))
    else:
        query_h = layer.local_forward(query_h)

```

Oracle selection is intentional. Training a router, materializer, and consumer at once would make a negative output result uninterpretable. Evidence masks are used only after the forward pass to divide final-token attention into evidence and distractor mass. They never enter model input or loss.

4 Controlled Experiment

4.1 Task and matched architectures

Each example draws fresh entity and relation symbols, builds a path of one to four facts, adds relation-sharing and label-relation distractors, and asks for the final entity. Stable entity-to-answer memorization is therefore unavailable. Facts are independently addressable references. The complete parent trains matched GlobalSA and LocalSA models; PRA receives a short query as direct context and selected facts as external K/V.

All models have 64 hidden dimensions, eight decoder layers, four heads, RoPE, and a 256-token operation limit. LocalSA uses a 16-token window. PRA replaces attention at zero-indexed layers 3 and 7, leaving LocalSA/FFN assimilation between memory interventions. This model is far below the nominal $< 100\text{M}$ Tier-0 ceiling: the goal is mechanistic discrimination at low cost.

4.2 Adaptation ladder

GlobalSA and LocalSA use identical random initial weights within each seed; only their attention windows differ. Every converted PRA regime starts from that seed’s trained LocalSA checkpoint.

1. **Frozen:** conversion only; no trainable parameters.
2. **Consumer LoRA:** low-rank updates to PRA Q/O and the immediately following FFNs.
3. **Interface LoRA:** Consumer LoRA plus PRA K/V.
4. **Broad LoRA:** low-rank updates to every decoder linear module.
5. **Full weight:** all converted-model parameters continue training.
6. **Native scratch:** the same PRA topology is present from initialization and all weights train end to end.

LoRA rank is eight with scale $\alpha/r = 2$. Baselines train for 800 steps; converted trainable regimes continue for 500; native scratch trains for the matched total of 1,300. We use five fixed model seeds, 4,096 deterministic training examples, and 512 held-out examples. The objective is answer-token cross entropy. No held-out result selects a checkpoint.

Regime	Trainable	Margin gain	ΔNLL	Selectivity
Frozen	0.000	1.480 ± 0.271	-0.197	0.332
Consumer LoRA	0.031	3.307 ± 0.191	-0.656	0.329
Interface LoRA	0.040	3.924 ± 0.456	-0.754	0.330
Broad LoRA	0.142	4.957 ± 0.318	-0.891	0.333
Full weight	1.000	6.470 ± 0.259	-1.062	0.333
Native scratch	1.000	10.992 ± 0.493	-1.583	0.343

Table 1: Five-seed controlled adaptation ladder. Trainable is the parameter fraction exposed during adaptation; ΔNLL compares evidence-only with the whole parent; selectivity is evidence’s share of whole-parent memory attention.

4.3 Causal interventions and metrics

For answer token y and final logits z , define the margin

$$m = z_y - \max_{v \neq y} z_v. \quad (5)$$

The primary usefulness effect compares correct and equally sized incorrect memory:

$$G_m = m_{\text{evidence}} - m_{\text{matched distractor}}. \quad (6)$$

We also report no-memory, evidence-only, and whole-parent conditions. Memory modularity is measured by

$$\Delta\text{NLL} = \text{NLL}_{\text{evidence}} - \text{NLL}_{\text{parent}}, \quad (7)$$

with analogous accuracy and margin differences. Evidence selectivity is evidence attention mass divided by total memory attention mass in the whole-parent condition.

To test whether memory objects become less dependent on neighboring facts, we encode the same evidence tokens alone and in the whole parent. At PRA-facing layer ℓ , key portability is

$$D_K^{(\ell)} = 1 - \cos\left(K_{e,\text{alone}}^{(\ell)}, K_{e,\text{parent}}^{(\ell)}\right), \quad (8)$$

averaged over heads, tokens, examples, and seeds; values are measured likewise.

5 Results

5.1 Adaptation capacity and causal usefulness

Table 1 reports the complete ladder. “Margin gain” is G_m ; its uncertainty column is the across-seed standard deviation. The best paired improvement over Frozen is 9.512 for Native scratch. This comparison is more diagnostic than evidence-versus-no-memory because it controls the number of imported tokens. Accuracy supplies the same practical picture. Frozen reaches 0.303 on evidence-only memory. Consumer, Interface, Broad, Full, and Native reach 0.821/0.875/ 0.948/0.964/ 0.992, respectively. Thus even the narrowest trainable consumer scope recovers a large part of the oracle-memory ceiling, while additional capacity continues to increase margin and confidence.

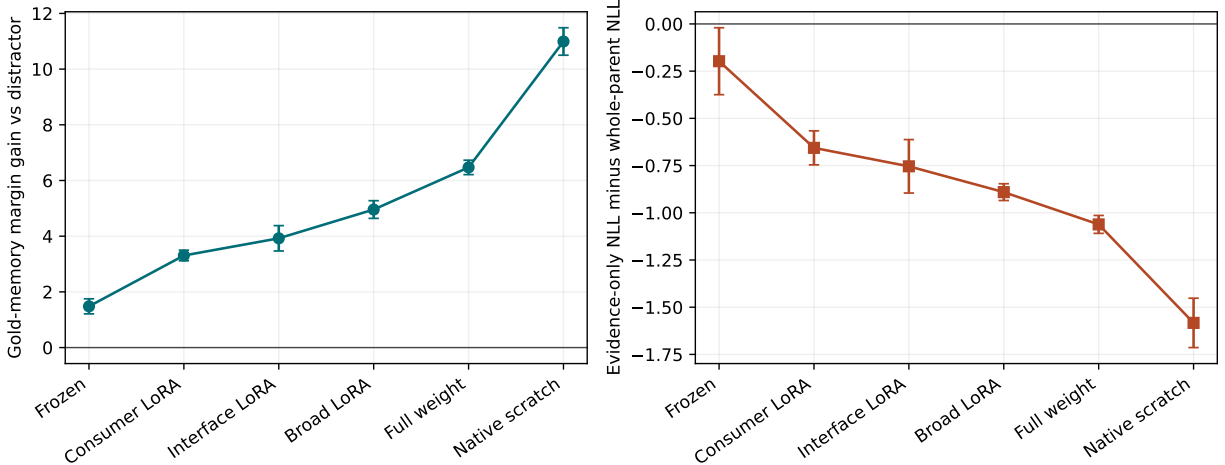


Figure 1: Left: answer-margin benefit of exact evidence over matched distractor memory. Right: evidence-only versus whole-parent NLL. Error bars are across-seed standard deviations, not confidence intervals.

5.2 Memory modularity and layer specialization

Whole-parent accuracy also improves: Frozen reaches 0.291, while Consumer, Interface, Broad, Full, and Native reach 0.481/0.495/ 0.480/0.484/ 0.477. This is a real noisy-memory gain, but it is much smaller than the evidence-only gain. No regime reduces the absolute evidence-versus-parent NLL gap; the least unfavorable change is -0.459 under Consumer LoRA. Gate 0 therefore rejects the strong modularity hypothesis. Whole-parent attention provides a stricter view: an adapted consumer should allocate its fixed memory mass to evidence rather than merely increase attention to every imported token. The strongest paired selectivity change is 0.011 for Native scratch. It has the same direction in all seeds but is small in magnitude. Adaptation mainly changes what the network can do with clean memory; it does not yet create a strong token-level filter among distractors.

5.3 Retention and chain depth

Converted LoRA models largely retain the original no-memory full-context path. Accuracy is 0.400 Frozen, 0.391 Consumer, 0.391 Interface, and 0.383 Broad. Full-weight continuation reaches 0.374. Native scratch reaches only 0.184 because it was trained through the sparse-memory path rather than the original full-context interface. The strongest oracle ceiling is therefore not the safest direct replacement for a pretrained model.

Evidence-only accuracy remains high through the four-hop stratum: Full and Native both reach 0.995 and 0.995, respectively. We do not interpret the upward shape of some curves as a causal depth benefit. Each stratum is a different randomized sample, and the generator was designed to prevent label leakage rather than calibrate equal difficulty by depth.

5.4 Representation portability

Figure 4 measures a different property from output quality. A low value means the same evidence token produces similar keys when encoded alone or amid distractors. A regime can improve answer use without making keys portable, or make keys portable without teaching downstream layers to

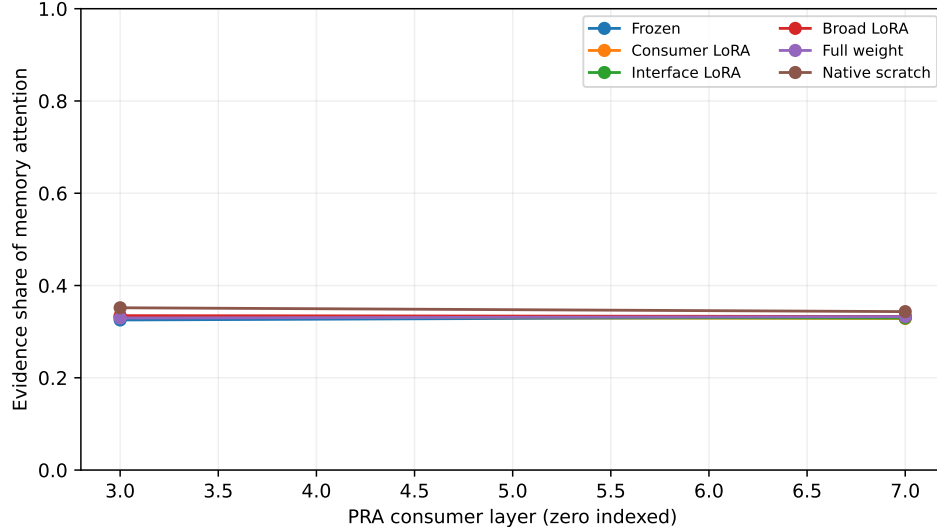


Figure 2: Evidence share of final-token memory attention at each PRA consumer. Evidence labels are diagnostic only and were not available to the model.

interpret values. Reporting both prevents an output-only explanation from being mistaken for a representation result.

5.5 Gate decision

The native-scratch evidence-memory accuracy minus LocalSA full-context accuracy is 0.592 on average. In total, 3 preregistered signals pass, producing `pass_to_tier1`. With five paired seeds, these results remain descriptive: even a same-direction effect in all seeds has a minimum exact two-sided sign-flip $p = 0.0625$. We therefore use the gate to decide the next experiment, not to claim a population-level scaling law.

6 Tier 1: Pretrained Ordinary-Language Gate

The first pretrained tier uses the 135M-parameter SmolLM2 checkpoint and the WikiText-2 reference corpus. Oracle evidence identity remains fixed, preserving the separation between consumption and discovery. Each of five deterministic train/validation splits runs 100 steps with ordinary full-context rehearsal every fourth step and evaluates 32 held-out continuations under no memory, matched distractor, selected evidence, whole-parent memory, and ordinary visible-context conditions. Consumer LoRA exposes 483,840 parameters at one PRA slot per four blocks; Interface LoRA adds K/V projections for 569,856 trainable parameters.

Both adaptation scopes optimize successfully. Mean evidence NLL falls by 0.618 for Consumer and 0.636 for Interface, while ordinary visible-context NLL also improves rather than regresses. The differentiable native-memory path is active rather than nominal: mean maximum K/V gradient norm is zero for Consumer by construction and 6.837 for Interface. Query/output and MLP gradients are nonzero in both regimes.

The stricter causal gate does not yet pass. Evidence has lower NLL than the matched distractor in four of five seeds for both adaptation scopes, with mean margins of 0.0468 and 0.0472 nats. On seed 71 the signs reverse slightly (-0.0128 and -0.0154). The large before/after gain therefore

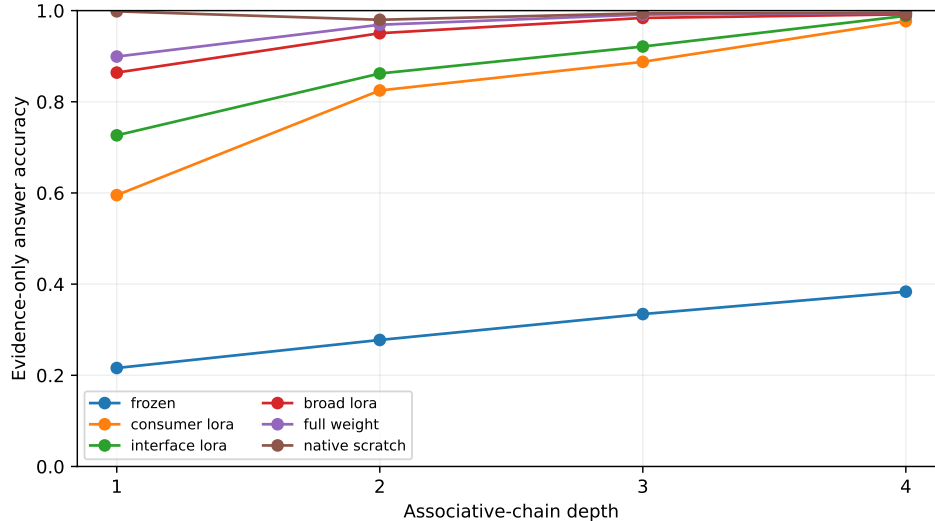


Figure 3: Evidence-only accuracy by associative-chain depth. Adaptation gains persist through depth four; cross-depth slopes are descriptive because difficulty is not calibrated across strata.

Table 2: Five-seed pretrained Tier-1 gate. Evidence NLL is the post-adaptation held-out mean; Δ NLL is relative to the same checkpoint before adaptation. Retention Δ is the ordinary visible-context NLL change, so negative values are improvements. Causal margin is matched-distractor NLL minus evidence NLL.

Regime	Trainable	Evidence NLL	Δ NLL	Retention Δ	Causal margin	Positive seeds
Frozen PRA	0	4.361	0.000	0.000	0.052	4/5
Consumer LoRA	483,840	3.743	-0.618	-0.563	0.047	4/5
Interface LoRA	569,856	3.724	-0.636	-0.577	0.047	4/5

includes general continuation learning and cannot by itself establish replicated reference-specific consumption. With five seeds, a same-direction result would in any case have a minimum exact two-sided sign-flip $p = 0.0625$. Learned routing remains disabled until a harder causal task produces a positive matched-distractor margin in every preregistered split.

7 Related Work

Sparse-attention methods reduce which token pairs interact inside a sequence [1, 10, 8]. Recurrent and compressed-memory transformers carry state across segments [3, 7]; Memorizing Transformers retrieve stored representations [9]. PRA differs in separating compact routing metadata from full-detail, layer-native K/V materialized under a bounded operation. This paper studies the learning interface after selection rather than proposing another retrieval score.

Retrieval-augmented language models train retrieval and generation jointly or in coordinated stages [4, 6, 2]. Our oracle stage is intentionally simpler. It provides a causal upper-bound question: if the correct physical memory is already present, can adaptation make it useful? A negative answer would make router optimization premature; a positive answer justifies reconnecting learned routing at the next gate.

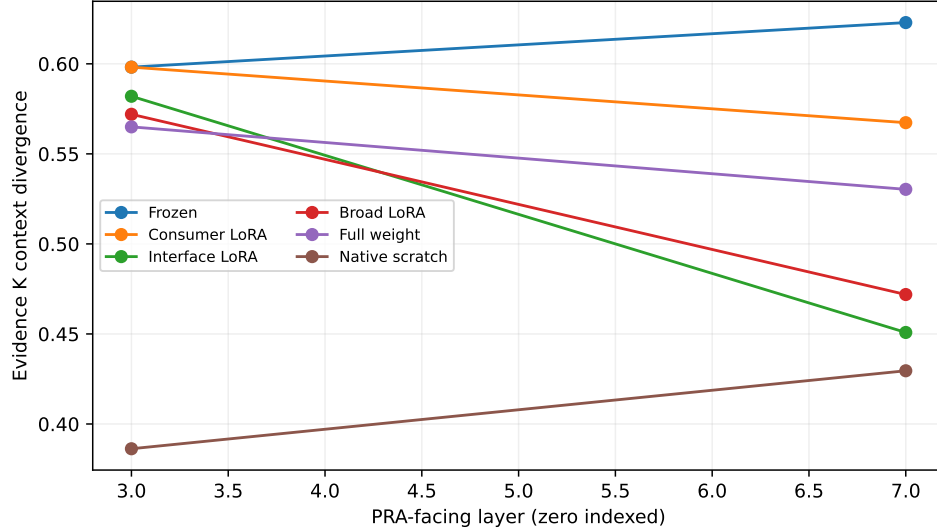


Figure 4: Context dependence of evidence keys at PRA-facing layers. Values near zero indicate more portable representations across evidence-only and parent encodings.

8 Limitations and Next Gate

Tier 0 uses a controlled vocabulary, answer-only objective, sub-million-parameter model, and short schedule. Tier 1 adds pretrained ordinary text, but remains a short continuation experiment rather than broad language-model evaluation. Oracle memory removes identity errors in both tiers and does not test top- k recall, graph traversal, or materialization budgets. The reported attention masses are correlates of use; the matched-distractor margin is the stronger causal output control. We do not measure production throughput because references remain in the training graph.

Tier 1 adds an ordinary-language objective and a pretrained model below 0.5B parameters. It validates the optimization path but misses the stricter all-seed causal criterion: both Consumer and Interface LoRA produce a positive evidence-versus-distractor margin in four of five splits. The next experiment therefore hardens the continuation task and repeats these two scopes before learning a router. If that causal margin replicates, routing advances with shuffled, disabled, and oracle controls. Native scratch remains a ceiling and a separate architecture-training question. Gemma-scale continuation remains a later gate, not a direct consequence of either the toy or 135M result.

9 Conclusion

Sparse memory has two learning problems: finding the right state and making that state computationally useful. By fixing the first, this study measures the second directly. The differentiable native-KV path and matched adaptation ladder turn a previously ambiguous frozen result into a testable training question. The toy outcome and pretrained optimization result support a stage-gated program, but the adverse causal split shows why optimization gain is not the same as reference-specific learning. Consumer plasticity must clear that causal gate before router complexity and model scale are increased.

References

- [1] Iz Beltagy, Matthew E. Peters, and Arman Cohan. Longformer: The long-document transformer. *arXiv preprint arXiv:2004.05150*, 2020.
- [2] Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George B. M. van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, et al. Improving language models by retrieving from trillions of tokens. In *International Conference on Machine Learning*, 2022.
- [3] Zihang Dai, Zhilin Yang, Yiming Yang, Jaime Carbonell, Quoc V. Le, and Ruslan Salakhutdinov. Transformer-XL: Attentive language models beyond a fixed-length context. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2978–2988, 2019.
- [4] Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. Retrieval augmented language model pre-training. In *International Conference on Machine Learning*, 2020.
- [5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.
- [6] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, 2020.
- [7] Jack W. Rae, Anna Potapenko, Siddhant M. Jayakumar, and Timothy P. Lillicrap. Compressive transformers for long-range sequence modelling. In *International Conference on Learning Representations*, 2020.
- [8] Aurko Roy, Mohammad Saffar, Ashish Vaswani, and David Grangier. Efficient content-based sparse attention with routing transformers. *Transactions of the Association for Computational Linguistics*, 9:53–68, 2021. Preprint: <https://arxiv.org/abs/2003.05997>.
- [9] Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations*, 2022.
- [10] Manzil Zaheer, Guru Guruganesh, Avinava Dubey, Joshua Ainslie, Chris Alberti, Santiago Ontanon, Philip Pham, Anirudh Ravula, Qifan Wang, Li Yang, and Amr Ahmed. Big bird: Transformers for longer sequences. In *Advances in Neural Information Processing Systems*, 2020.